LFM2.5-VL-450M · Liquid AI

EXECUTIVE BRIEF · FOR THE VISUAL-DATA BUYING COMMITTEE

The vision model that belongs in your data pipeline — not your product.

A 450M-parameter multimodal model, small and cheap enough to run over your **entire image corpus** on hardware you already own, over data that never leaves your network — and to re-run it, free, every time your needs change.

Who this is for: the team that owns visual data infrastructure at a Fortune-2000 — millions of claims photos, product shots, inspection frames, or media assets that need captioning, tagging, detection, and triage at scale, and a budget that re-processing keeps inflating.

01 · OWN THE LINE ITEM

Re-processing becomes free

A pipeline re-reads the whole corpus on every model or taxonomy change. Metered APIs charge for all of it, again. On-device marginal cost is **≈ $0**.

02 · KEEP DATA HOME

Nothing leaves the boundary

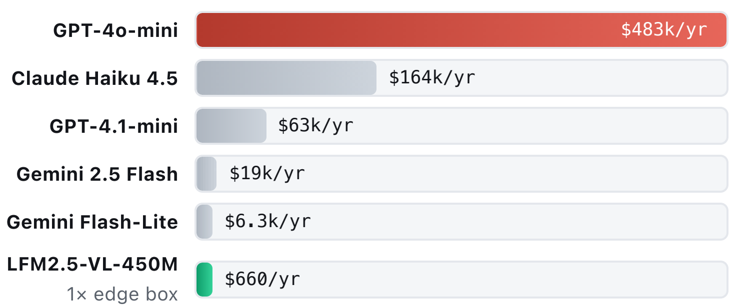
Runs fully on-prem / at the edge. Point it at regulated or sensitive imagery a cloud API is forbidden to see. **Zero egress.**

03 · OWN A SPECIALIST

Fine-tune to your taxonomy

Small enough to fine-tune on **one GPU**, then freeze. No vendor drift, no rate limits, no lock-in. The weights are yours.

COST TO RUN + RE-RUN A 30M-IMAGE ARCHIVE · PER YEAR



Representative: 30M corpus, 3M new/mo, re-processed 3×/yr = 126M inferences. API = input tokens only, 2026 list price. On-device = one edge box, hardware amortized. Marginal cost per re-run: $0.

THE FIELD, HONESTLY

OPTION	RUNS	$/1K IMG	DATA HOME
LFM2.5-VL-450M	edge·CPU·GPU	≈$0 mrg	Yes
Qwen2.5-VL-3B (self-host)	GPU, heavier	≈$0 mrg	Yes
Gemini Flash-Lite (API)	cloud only	$0.05	No
GPT-4o-mini (API)	cloud only	$3.83	No
Silicon-partner VLM	one NPU SKU	≈$0 mrg	Yes

Qwen2.5-VL-3B scores higher on general VQA but is ~6× the size; the APIs are most capable yet meter every image and require egress.

Where it is not the answer: dense text extraction (OCRBench 684 vs Qwen-7B's 864). Route document transcription to a specialist or larger model — this model wins on detection, captioning, counting, and routing, the triage layer. **License:** LFM Open License v1.0, not Apache/MIT — legal review before fleet deployment.

The ask: a two-week test built to fail fast if it's wrong

Take 100k images from one corpus → fine-tune LFM2.5-VL-450M on your taxonomy (single GPU, hours) → run locally → measure triage precision/recall and cost-per-million vs your current API or human-labeling baseline. Miss the bar: you spent a GPU-week. Clear it: you removed a compounding re-processing cost and a data-governance blocker at once.